

Tutorial Exercises - Week 1

4001CMD - Mathematical Skills for Computing Professionals

The following exercises should be attempted before your next tutorial session this week. They focus on material covered in Week 1, namely set theory and 1st part of symbolic logic.

Set Theory

Problem 1. For the following, show rigorously that $A = B$.

- (a) $A = \{1, 2, 3\}$, $B = \{n : n \in \mathbb{N}, n^2 < 10\}$.
- (b) $A = \{x \in \mathbb{R} : x^2 - 4x + 4 = 1\}$, $B = \{1, 3\}$

Problem 2. For the following, show rigorously that $A \neq B$.

- (a) $A = \{1, 2\}$, $B = \{x \in \mathbb{R} : x^3 - 2x^2 - x + 2 = 0\}$
- (b) $A = \{1, 3, 5\}$, $B = \{n : n \in \mathbb{N}, n^2 - 1 \leq n\}$

Problem 3. For the following, determine whether $A = B$.

- (a) $A = \{x : x^2 + x = 2\}$, $B = \{1, -1\}$
- (b) $A = \{x \in \mathbb{R} : 0 < x \leq 2\}$, $B = \{1, 2\}$

Problem 4. For the following, determine whether that $A \subseteq B$.

- (a) $A = \{1, 2\}$, $B = \{x : x^3 - 6x^2 + 11x = 6\}$
- (b) $A = \{2n : n \in \mathbb{N}\}$, $B = \mathbb{N}$
- (c) $A = \{x \in \mathbb{R} : x^3 - 2x^2 - x + 2 = 0\}$, $B = \{1, 2\}$
- (d) $A = \{1, 2, 3, 4\}$, $C = \{5, 6, 7, 8\}$, $B = \{n \in \mathbb{N} : n + m = 8 \text{ for some } m \in C\}$

Problem 5. The *symmetric difference* for two sets A and B is the set defined by

$$A \triangle B = (A \cup B) \setminus (A \cap B)$$

(recall for sets X and Y with $Y \subseteq X$, $X \setminus Y$, read “ X minus Y ”, is the set of elements of X that are not elements of Y).

- (a) If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4, 5\}$, find $A \triangle B$.
- (b) Describe in words the symmetric difference of two sets A and B (not necessarily those sets in part (a)).

Symbolic Logic 1

Problem 6. Given that p is false, q is true, r is false, and s is true, determine whether the following propositions are true or false.

- (a) $\neg p \vee \neg(q \wedge r)$
- (b) $\neg(p \vee q) \wedge (\neg p \vee r)$
- (c) $(p \vee \neg r) \wedge \neg((q \vee r) \vee \neg(r \vee p))$
- (d) $(p \Rightarrow q) \wedge (q \Rightarrow r)$
- (e) $(p \Rightarrow q) \Rightarrow r$
- (f) $p \Rightarrow (q \Rightarrow r)$
- (g) $(s \Rightarrow (p \wedge \neg r)) \wedge ((p \Rightarrow (r \vee q)) \wedge s)$
- (h) $((p \wedge \neg q) \Rightarrow (q \wedge r)) \Rightarrow (s \vee \neg q)$

Problem 7. For each of the following, formulate the symbolic expression in words, assuming the propositions stand for the following statements:

p : You play football.
 q : You miss the midterm exam.
 r : You pass the module.

- (a) $p \vee q \vee r$
- (b) $\neg(p \vee q) \vee r$
- (c) $(p \wedge q) \vee (\neg q \wedge r)$

Problem 8. For each of the following, represent the proposition symbolically by letting

p : You run 10 laps daily.
 q : You are healthy.
 r : You take multi-vitamins.

- (a) You run 10 laps daily, but you are not healthy.
- (b) You do not run 10 laps daily, you do not take multi-vitamins, and you are not healthy.
- (c) Either you are healthy or you do not run 10 laps daily, and you do not take multi-vitamins.